

Automatic detection of the fetal brain midsagittal plane on MRI using a deep learning pipeline



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BACKGROUND: Accurate localization of the midsagittal plane (MSP) is essential for evaluating midline brain structures such as the corpus callosum and posterior fossa on fetal MRI. However, this task is often challenging due to fetal motion and the relatively large slice thickness of conventional acquisitions. Automated MSP detection could enhance image quality evaluation and promote a more reliable and standardized assessment of the fetal brain anatomy.

OBJECTIVE: To develop an automated deep learning pipeline for automatic fetal brain MSP detection from fetal brain MRI scans

METHODS: We retrospectively included fetuses with normal brain anatomy between 18 and 36 weeks who underwent routine 1.5 T fetal brain MRI using single-shot fast spin-echo T₂-weighted sequences. Ground-truth MSP were annotated by an expert in fetal MRI, and cases were labeled as MSP-present or MSP-absent. A multitask 2D U-Net was trained to identify the MSP by detecting the concurrent visibility of 4 midline structures: the genu and splenium of the corpus callosum, the cerebellar vermis, the pons, and to generate structure-specific probability maps and slice-level MSP scores, defined as the predicted probability that each sagittal slice corresponds to the true MSP. These were aggregated by a LightGBM classifier to produce a case-level decision. Model performance

was assessed using 5-fold cross-validation, with 80% of cases used for training and 20% for testing in each iteration. Evaluation metrics included area under curve (AUC), accuracy, sensitivity, specificity, and localization accuracy (distance between predicted and expert-annotated MSP).

RESULTS: A total of 432 fetal brain MRI stacks from 225 fetuses were included at 32 (interquartile range, 28–34) weeks, of which 135 contained an identifiable MSP and 297 did not. The model achieved excellent slice-level performance to detect MSP, with an AUC of 0.95 and an accuracy of 0.87. At the case level, the AUC and accuracy were 0.81 and 0.74, respectively. In 99% of true-positive cases, the predicted MSP was localized within ± 1 slice of the expert annotation, and in 75% of cases, the correspondence was exact.

CONCLUSION: The proposed deep learning framework accurately and consistently identifies the fetal brain MSP on routine MRI. Automated localization of the MSP offers an objective quality control measure, facilitating more accurate anatomical and biometric evaluations of the fetal brain.

Key words: fetal brain MRI, midsagittal plane detection, deep learning, automated image analysis, corpus callosum

Introduction

Prenatal magnetic resonance imaging (MRI) has become an essential complement to ultrasound for the assessment of fetal brain development, particularly when ultrasound findings are inconclusive or structural anomalies are suspected.^{1–4} Among the standard anatomical plane, the midsagittal plane (MSP) provides the optimal view for assessing midline structures such as the corpus callosum, cerebellar vermis and brainstem.^{5–7} Therefore, precise identification of the MSP is important for accurate anatomical evaluation and reliable

biometric measurements, in particular corpus callosum anomalies.^{5,8,9}

However, obtaining an exact midsagittal slice is often challenging in fetal MRI. Fetal motion, variable head orientation, and the relatively large slice thickness of conventional acquisitions frequently lead to suboptimal plane positioning, compromising both the visibility of midline structures and the reproducibility of morphometric measurements.^{10–14} This limitation not only affects diagnostic accuracy but also prevents objective quality control and the establishment of standardized biometric references. An automated approach capable of reliably detecting the MSP could therefore improve the quality of fetal MRI examinations by providing real-time feedback to the operator or enabling retrospective quality assessment of acquired datasets.

Recent advances have addressed several downstream tasks in fetal brain MRI analysis, including multi-tissue brain segmentation,¹⁵ automated linear

biometry,¹⁶ and corpus callosum segmentation with downstream biomarker extraction.¹⁷ However, for workflows that rely on standardized midline measurements or sagittal biometric assessment, reliable identification of the midsagittal plane is mandatory. While several automated methods have been developed for fetal image analysis,¹⁸ the specific task of MSP detection has not yet been addressed. For instance, Sanchez et al.¹⁹ proposed FetMRQC, an automated quality-assessment tool that operates at the volume level, providing a global image-quality score but lacking slice-level localization and anatomical interpretability. In fetal ultrasound, deep learning approaches can successfully recognize standard planes,^{20,21} but these methods are modality-specific and not directly transferable to MRI due to fundamental differences in image. Acquisition-based approaches can assist with prospective slice prescription but require dedicated sequences and are not applicable to retrospective clinical data.²²

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AJOG MFM at a Glance

Why was this study conducted?

Accurate identification of the fetal brain midsagittal plane is essential for assessing midline structures but is frequently unreliable on fetal MRI due to motion and slice thickness. An automated, objective solution was needed to improve reproducibility and quality control.

Key findings

the proposed deep learning pipeline accurately detects the midsagittal plane with excellent slice-level performance and robust case-level localization. In almost all true-positive cases, the predicted plane was within one slice of expert annotation.

What does this study add to what is known?

This is the first MRI-specific, anatomically grounded deep learning method for midsagittal plane detection. It provides an objective, interpretable tool that enables standardized anatomical assessment and supports large-scale, reproducible fetal brain MRI studies.

The objective of this study was to develop and validate an automated deep learning pipeline for accurate detection of the fetal brain MSP on routine clinical fetal MRI scans.

Methods**Study design and population**

We conducted a retrospective, single-center study at the LUMIERE platform, Necker-Enfants Malades Hospital & University of Paris, Paris, France. In the Deep learning chapter: "the full source code, trained models, and documentation are publicly available at https://github.com/rocorroenne/MSP_detection", between 12/2019 and 04/2024. We included women between 22 and 36 weeks' gestation who have agreed to participate to our fetal MRI research program, dedicated to advancing prenatal imaging research to improve maternal and fetal health. We excluded fetuses with known congenital infections, structural or chromosomal abnormalities, as well as cases affected by motion artifacts, technical issues, or errors during Digital Imaging and Communications in Medicine (DICOM) to Neuroimaging Informatics Technology Initiative (NIfTI) data conversion. All participants provided written informed consent approved by the Institutional Review Board (2019-A01498-49).

MRI acquisition

All examinations were performed on a GE 1.5 T MRI scanner (GE Healthcare, Chicago, IL). Sagittal images were acquired using a single-shot fast spin echo (SSFSE) T₂-weighted sequence with typical parameters: repetition time (TR) $\approx$ 1.8 seconds; echo time (TE) $\approx$ 120 milliseconds; flip angle=90°; in-plane resolution=0.5859 $\times$ 0.5859 mm; slice thickness=4 mm; and no interslice gaps. No motion correction or super-resolution methods were used.

Image preprocessing

All MRI volumes were converted from DICOM to NIfTI format using dcm2niix (v1.0.20241211).²³ A brain mask was automatically generated for each stack using MONAIbbs.²⁴ All subsequent preprocessing and model training were performed exclusively within the masked brain region to exclude extracranial tissues and background noise. Detailed preprocessing procedures, including brain masking, intensity normalization, and affine preservation policies, are provided in the [Supplementary Material](#).

Ground truth definition

The MSP was defined as the sagittal slice in which the genu and splenium of the corpus callosum, the cerebellar vermis, and the pons were simultaneously visible ([Figure 1](#)).^{3,5} Each MRI stack was categorized as MSP-present when such a slice was identifiable, or MSP-absent when it

was not. All annotations were performed using ITK-SNAP²⁵ by a maternal–fetal medicine specialist with more than 5 years of experience in fetal MRI. The morphological preprocessing steps used for ground-truth generation are detailed in the [Supplementary Material](#).

Deep learning model

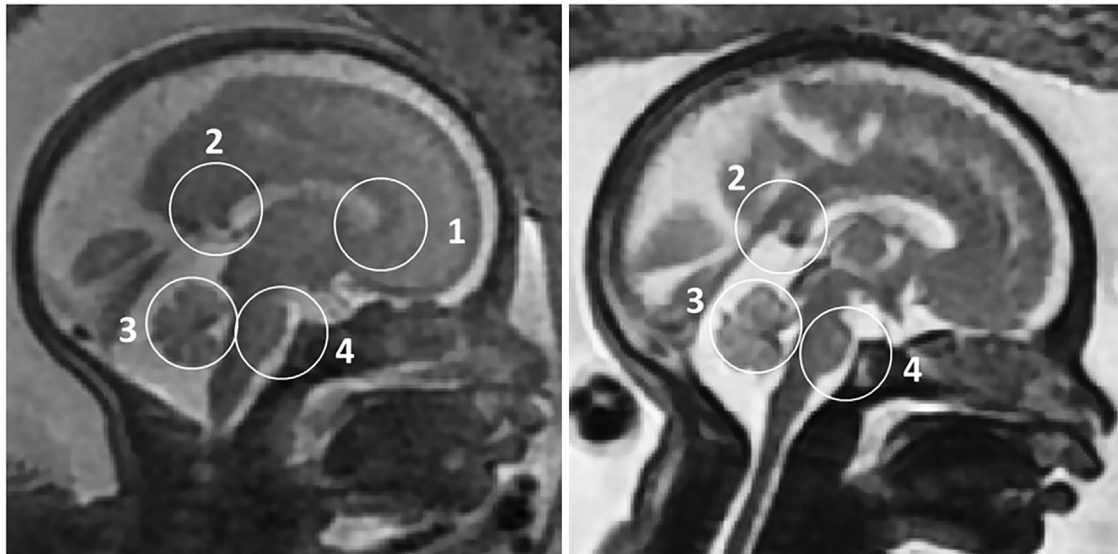
A multitask 2D U-Net architecture was implemented to simultaneously generate structure-specific probability maps (heatmaps) for the genu and splenium of the corpus callosum, the cerebellar vermis, and the pons, and to predict the slice-level probability of MSP presence. A 2D design was chosen over a 3D network to account for variable slice thickness and motion-related inconsistencies across fetal MRI stacks. The slice-level predictions were then aggregated using a LightGBM classifier²⁶ trained on quantitative features extracted from the predicted heatmaps to produce a final case-level decision. The LightGBM classifier operates on individual slices without incorporating information from adjacent slices. Through-plane context is only indirectly accounted for at the case level, where the final decision is based on the highest-scoring slice. A schematic representation of the pipeline is shown in [Supplementary Figure 1](#). The full network architecture, loss functions, and 2-stage training strategy are detailed in the [Supplementary Material](#). A video representing one successful case and one unsuccessful case is provided in [Supplementary Video 1](#). The full source code, trained models, and documentation are publicly available at https://github.com/rocorroenne/MSP_detection.

Evaluation and statistical analysis

Model performance was evaluated at both the slice and case levels.

At the slice level, all sagittal slices were pooled across subjects to assess the model's ability to discriminate between midsagittal and non-midsagittal slices. Performance metrics included the area under the receiver operating characteristic curve (AUC), accuracy, sensitivity, specificity, and F1 score.

FIGURE 1

Sagittal T₂-weighted fetal brain MRI showing an MSP-present case (left) and MSP-absent case (right)

1-splenium, 2-genu, 3-vermis, 4-pons.

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At the case level, slice-level probabilities were aggregated using the LightGBM classifier to obtain a final classification per subject.²⁶ Case-level performance was quantified using AUC, accuracy, and localization accuracy, defined as the absolute slice-wise distance between the predicted and expert-annotated MSP, expressed in number of slices.

Five-fold GroupKFold cross-validation²⁷ was applied to prevent data leakage between subjects. Descriptive statistics are presented as median and interquartile range and compared using the Mann–Whitney U test. A *p*-value <.05 was considered statistically significant.

All analyses were performed in Python (version 3.10) using PyTorch,²⁸ scikit-learn,²⁹ LightGBM,²⁶ SciPy,³⁰ and Alumentations³¹ libraries. Additional details on feature extraction, meta-classifier design, and cross-validation strategy are provided in the [Supplementary Material](#).

Results

The final study cohort included 432 fetal MRI images from 225 fetuses, of which 135 were MSP-present and 297 MSP-absent ([Figure 2](#)). The median gestational age was 32 [IQR, 28–34]

weeks, with gestational age identical in the MSP-present (29 [25–32] weeks) and MSP-absent groups (28 [22–33] weeks, *p*=.69).

Slice-level performance

The model achieved excellent discrimination between MSP-present and MSP-absent slices (AUC=0.95 [95% CI 0.94–0.96]; accuracy=0.87 [0.84,0.90]; Sensitivity=0.70 [0.58–0.81] and specificity=0.96 [0.94–0.98]; [Figure 3A](#)). Similar performance was observed across all gestational age subgroups ([Supplementary Figure 2](#)). Predicted heatmaps accurately localized the corpus callosum, cerebellar vermis, and pons, showing strong spatial agreement with expert annotations ([Figure 4](#)).

Case-level performance

At the case level, the model demonstrated robust discrimination between MSP-present and MSP-absent cases (AUC=0.81 [0.77–0.85]; accuracy=0.74 [0.69–0.78]; sensitivity=0.81 [0.74–0.87]; specificity=0.70 [0.65–0.76]; [Figure 3B](#)). Similar performance was observed across all gestational age subgroups ([Supplementary Figure 1](#)). In

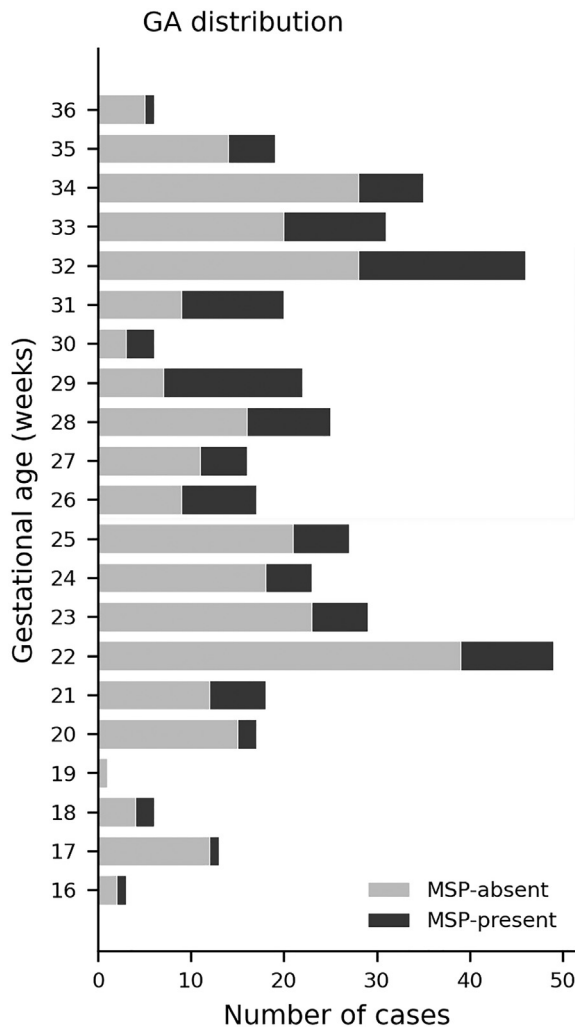
99% of true-positive cases, the predicted MSP was localized within ± 1 slice of the expert annotation, with a maximum physical distance of 4.0 mm given the slice thickness, and in 75% of cases the correspondence was exact (0 mm deviation), indicating excellent spatial agreement and consistent localization performance.

Discussion

In this study, we developed and validated a fully automated deep learning pipeline to detect the MSP in fetal brain MRI. The model demonstrated strong performance and localization accuracy, with most predicted planes falling within one slice of expert annotations. To our knowledge, this is the first MRI-specific deep learning approach designed for fetal MSP identification. Automated MSP localization could standardize anatomical assessment and biometric measurements, improving reproducibility and reducing observer variability in fetal brain analysis. Beyond anatomical standardization, automated MSP detection may also support real time quality feedback during acquisition. By detecting stacks where no true MSP is present, this approach could help operators to adjust

FIGURE 2

Distribution of MSP-present and MSP-absent slices across gestational age



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fetal head positioning or acquire additional views. Such functionality may improve the overall quality of fetal MRI examinations.

Existing quality-control methods in fetal MRI, such as FetMRQC,¹⁹ provides global image quality scores but lack anatomical interpretability and slice-level localization. In contrast, our framework performs anatomically informed, slice-level MSP detection while generating structure specific heatmaps for transparent and interpretable decision making. Although deep learning models have successfully achieved standard plane recognition in fetal ultrasound, their direct transfer to MRI is

limited by differences in image contrast, signal-to-noise ratio, and acquisition physics³². Similarly, acquisition based methods such as rapid head-pose detection may facilitate prospective slice prescription but do not enable retrospective quality control.²² Our approach thus represents the first fetal MRI-specific, anatomically grounded, and retrospective solution for reliable MSP localization. The absence of prior MRI specific solutions also means that current fetal brain studies frequently rely on subjective visual selection of the MSP, introducing methodological bias. By providing a standardized and objective definition, our framework could improve

the comparability of biometric studies on brain development.

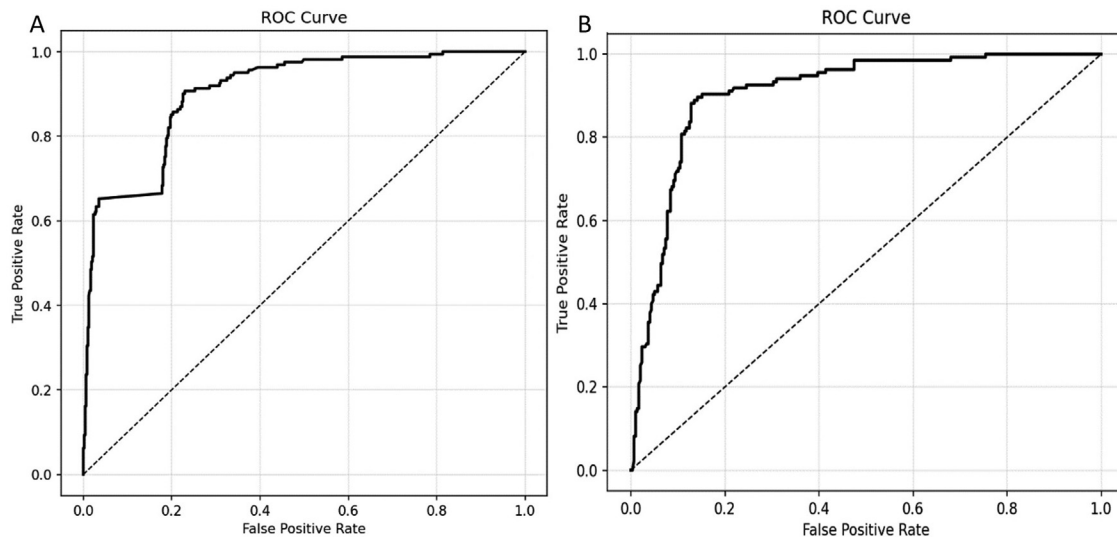
At the slice level, the network benefited from direct visual discrimination of midline anatomy, achieving consistent classification performance. The slight reduction in case-level metrics likely reflects residual fetal motion and partial visibility of structures across slices. Nevertheless, the model maintained consistent localization accuracy, underscoring its robustness under real-world acquisition conditions where motion and anatomical variability are inevitable.

Accurate identification of the MSP is essential for assessing midlines structures such as the corpus callosum and cerebellar vermis^{5,8}. Manual localization is inherently subjective, time-consuming, and susceptible to inter- and intra-observer variability. By providing an automated and objective definition of the MSP, our model has the potential to improve the reproducibility of biometric analyses, standardize measurements across institutions, and streamline clinical workflow by automatically highlighting diagnostically relevant slices. Beyond individual scans, such automation may also enable large-scale, multicenter studies of fetal brain development by ensuring methodological consistency across datasets.

This study has several methodological strengths. First, the study included a relatively large cohort of healthy fetuses between 22 and 36 weeks. Second, cross-patient validation was implemented to prevent data leakage and overfitting. Third, the definition of ground truth was strict, requiring the simultaneous display of 4 structures to ensure anatomical validity. Finally, the incorporation of heatmaps enhanced model transparency.

However, some limitations must be acknowledged. Firstly, it was conducted at a single center using a single MRI protocol, and external validation on datasets acquired by different scanners, field strength was not performed. Secondly, there was no motion correction or super-resolution applied because this model is designed to process raw and unprocessed sagittal stacks so as to

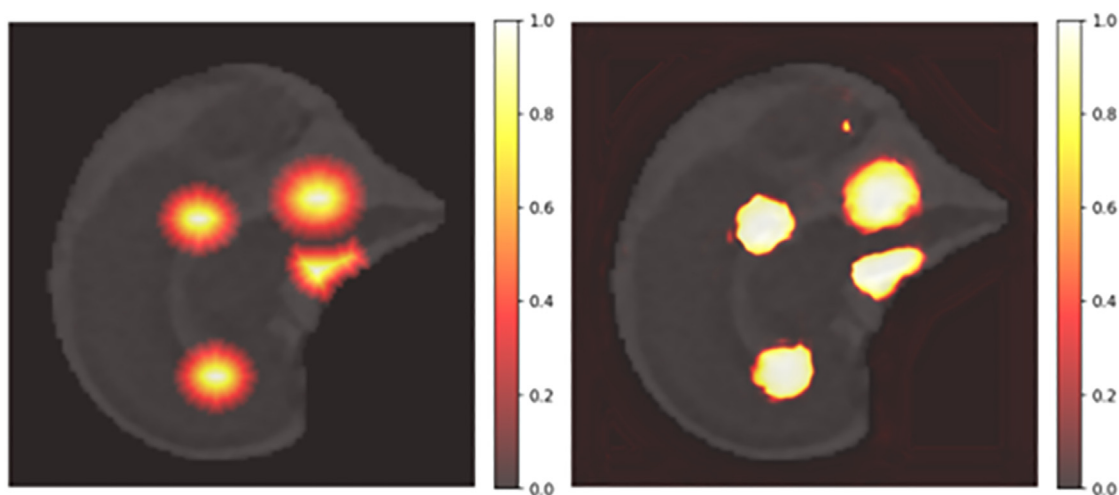
FIGURE 3
ROC curves for midsagittal plane detection



(A) Slice-level performance showing excellent discrimination between midsagittal and non-midsagittal slices (AUC=0.95). (B) Case-level performance after aggregation of slice predictions using the LightGBM classifier (AUC=0.81).

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FIGURE 4
Multi-structure localization performance



(A) Ground-truth (GT) heatmap showing the reference locations of the genu and splenium of the corpus callosum, the cerebellar vermis, and the pons, as annotated by the expert. Each structure was represented as a separate channel, and all channels were summed for visualization. (B) Predicted heatmap generated by the model, obtained by summing the 4 output channels corresponding to the same anatomical structures. The strong spatial overlap between the GT and predicted maps demonstrates that the model accurately identifies and localizes multiple midline structures within a single midsagittal slice.

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remains applicable regardless of whether reconstruction is performed. For centers that do employ motion correction and super-resolution

reconstruction, the proposed method could function as a preceding step within such pipelines, although this integration was not evaluated in the

present study and warrants future investigation. Thirdly, MSP annotation was performed by a single expert, precluding assessment of inter-observer

variability. In addition, due to the slice thickness, the MSP corresponds to a physical slab rather than an ideal geometric plane, introducing partial-volume effects. Nonetheless, the model localized the MSP within ± 1 slice of expert annotation in 99% cases, confirming its robustness within clinical imaging constraints. A distinction should be made between MSP detection and its suitability for diagnostic interpretation. While our model identifies the presence of the MSP, it does not ensure that image quality is sufficient for diagnostic assessment. Its role is therefore limited to providing an objective quality control step and facilitating standardized slice selection for biometric measurements. Future work should also address challenging anatomical conditions such as callosal agenesis or dysgenesis, where the genu or splenium may be absent or malformed, requiring alternative anatomical landmarks for accurate MSP estimation. Future studies may also benefit from 3D architecture or integration with motion correction algorithms, which could improve performance in severely motion-degraded datasets.

Conclusion

This study presents the first fetal MRI-specific deep learning framework for automated MSP detection in fetal brain imaging. The method combines high slice-level discriminative accuracy with clinically reliable case-level localization, providing an interpretable and reproducible solution to automatically detect MSP. By reducing observer dependency and standardizing the initial step of anatomical and morphometric analysis, this approach has the potential to improve diagnostic consistency in clinical practice and to facilitate large-scale, multi-center research on fetal brain development.

Clinical trials registry name and number

LUMIERE on the FETUS (FETUS), NCT04142606

CRedit authorship contribution statement

Zecheng Zhao: Writing — original draft, Formal analysis, Data curation, Conceptualization. **David Grevent:** Writing — review & editing, Methodology. **Sunniva Vaillant:** Writing — review & editing, Resources. **Mingen Xu:** Writing — review & editing. **Laurence Bussi res:** Writing — review & editing. **Laurent J. Salomon:** Writing — review & editing, Supervision, Methodology, Conceptualization. **Romain Corroenne:** Writing — review & editing, Writing — original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

Artificial intelligence tools (ChatGPT, OpenAI, San Francisco, CA) were used to assist in improving the grammar, clarity, and style of the manuscript. The authors reviewed and approved all content generated by the AI tool.

Supplementary materials

Supplementary material associated with this article can be found in the online version at [doi:10.1016/j.ajogmf.2026.101962](https://doi.org/10.1016/j.ajogmf.2026.101962).

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